

MOVIE RECOMMENDATION ENGINE USING MACHINE LEARNING

CodeTech IT Solutions Internship Project Documentation

Project Information

Project Name: Movie Recommendation Engine

Domain: Artificial Intelligence

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Duration: 6 Weeks

Organization: CodeTech IT Solutions

1. Introduction

Movie recommendation systems help users discover movies matching their interests. This project uses TF-IDF Vectorization and Cosine Similarity to recommend similar movies.

2. Project Objective

Implement a content-based recommendation system and recommend similar movies.

3. Project Scope

Load data, preprocess, vectorize text, calculate similarity and recommend movies.

4. Technologies Used

Python, Pandas, NumPy, Scikit-learn, Git and GitHub.

5. Dataset

TMDB 5000 Movie Dataset with about 4803 movies.

6. Technique

Content-Based Recommendation using TF-IDF and Cosine Similarity.

7. Methodology

Load dataset, clean data, create TF-IDF vectors, compute cosine similarity, take user input, recommend Top 10 movies, save recommendations.csv.

8. Folder Structure

CodeTech-Movie-Recommendation-Engine with data, src, outputs, screenshots, documentation, README.md and requirements.txt.

9. Output Summary

Dataset loaded successfully, TF-IDF vectors created, cosine similarity calculated, Top 10 recommendations generated and recommendations.csv saved.

10. Applications

OTT platforms, streaming services, entertainment systems and recommendation engines.

11. Advantages

Fast, simple, NLP-based and easily extendable.

12. Future Enhancements

Add genres, actors, directors, hybrid recommendation systems and web deployment.

13. Conclusion

The project successfully demonstrates a content-based Movie Recommendation Engine using Machine Learning, providing practical experience with NLP, TF-IDF and similarity-based recommendation.

References

TMDB 5000 Movie Dataset
Python Documentation
Scikit-learn Documentation
Pandas Documentation
NumPy Documentation